

Lab 3 – Architecture Selection Justification

Automated Rubric Assignment Evaluator

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Scenario: Automated Rubric Assignment Evaluator

Selected Architecture

Microservices Architecture is selected for the Automated Rubric Assignment Evaluator. The system is divided into independent services for submission handling, evaluation queuing, test execution, rubric scoring, and peer review. This separation allows individual services to be developed, deployed, and scaled independently.

Scenario-Specific Reason 1 – Scalability

The requirements specify support for up to 100 concurrent submissions without losing queue items or exceeding the memory limit. The Test Execution Service can be scaled independently by adding workers when the number of evaluation jobs increases, instead of scaling the complete system.

Scenario-Specific Reason 2 – Fault Isolation

The evaluation flow executes syntax and unit tests before generating the rubric score. A failed test or invalid submission should not stop other submissions in the queue. Separating test execution from scoring and peer-review services provides better fault isolation and allows the affected submission to be handled independently.

Security Advantage

Test execution is separated from the other evaluation services, reducing the impact of failures during compilation or test execution. In addition, the requirements specify AES-256 encryption at rest for stored score and personal data, with submissions, logs, and evaluation records retained for at least two academic years.

Performance Benefit

The Evaluation Queue can distribute jobs to available test-execution workers, allowing multiple submissions to be processed concurrently. This supports the concurrent-submission requirement and helps the system produce score breakdowns without making all evaluation work depend on a single execution process.

Component Interaction

The User & Evaluation Dashboard submits project ZIP files to the Submission Service. Jobs are passed through the Evaluation Queue to the Test Execution Service, whose test results are sent to the Rubric Scoring Service. The resulting evaluation information is stored in the Evaluation Data Store, while the Peer Review Service handles reviewer assignment and peer-review information.

Automated Rubric Assignment Evaluator Microservices Architecture

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